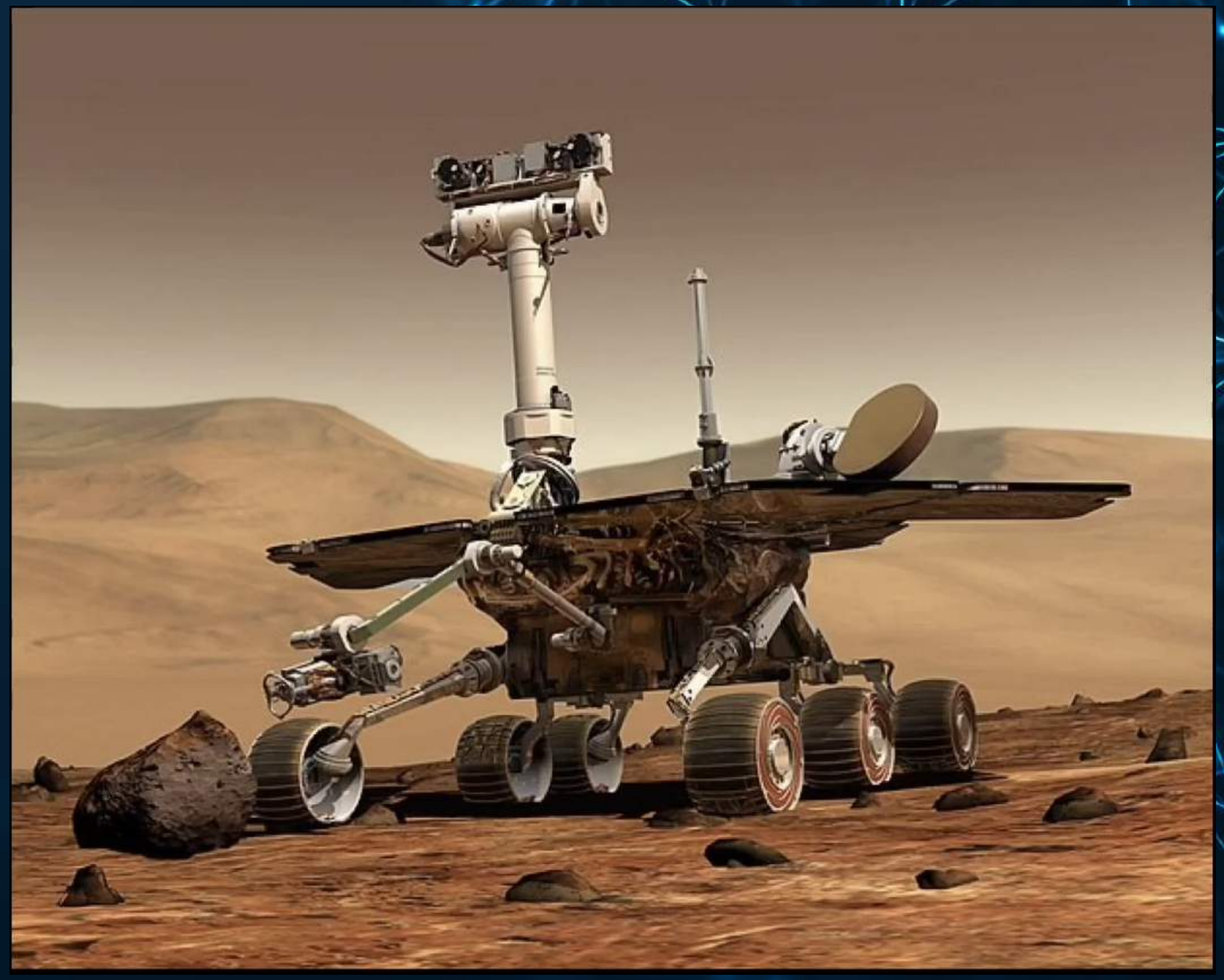
Incertezas

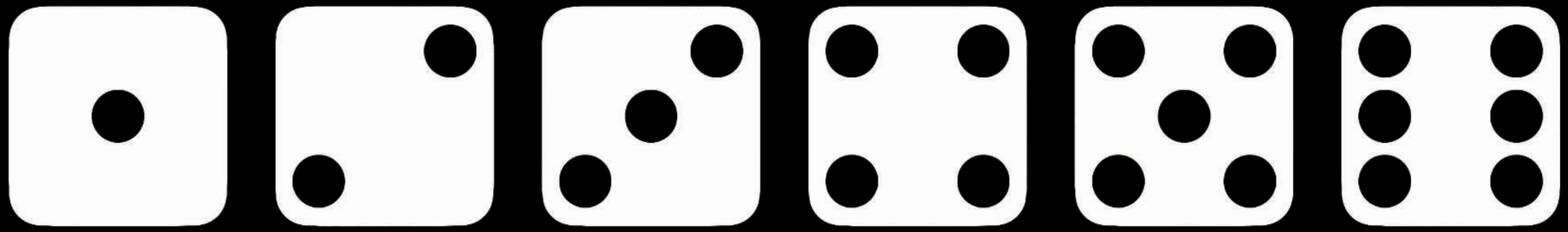
IA



NEXT 36 HOURS

HOURLY → | 10 DAYS →

TONIGHT	THU	THU NIGHT	FRI	FRI NIGHT
CLEAR				
LOW 20°	HIGH 36°	LOW 25°	HIGH 46°	LOW 32°
0%	0%	0%	0%	20%



Probabilidade

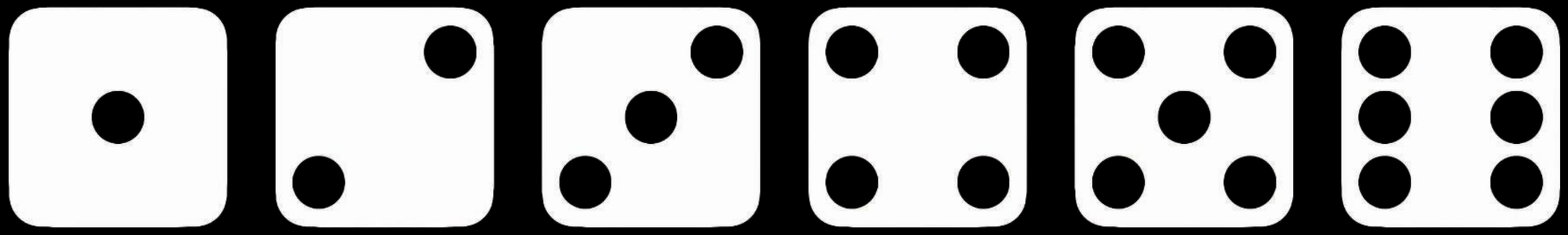
Mundos Possíveis

$$\omega$$

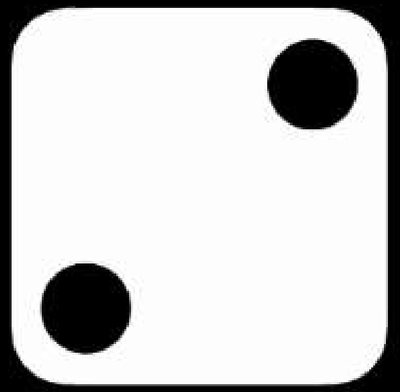
$$P(\omega)$$

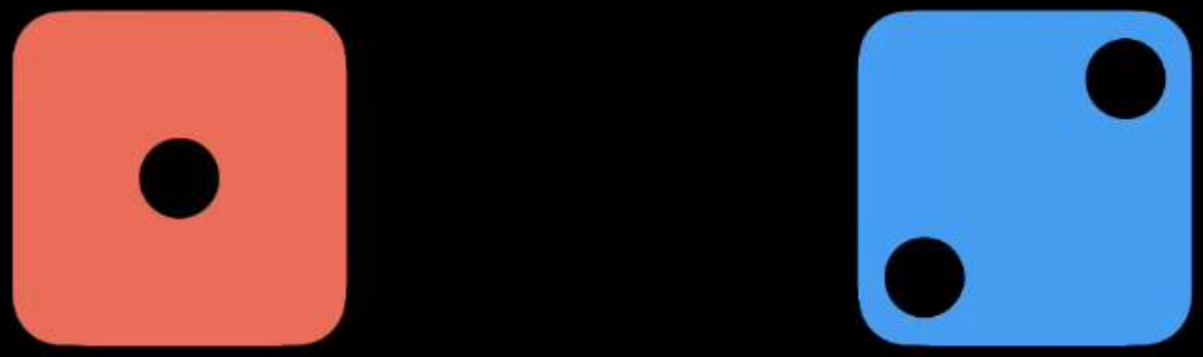
$$0 \leq P(\omega) \leq 1$$

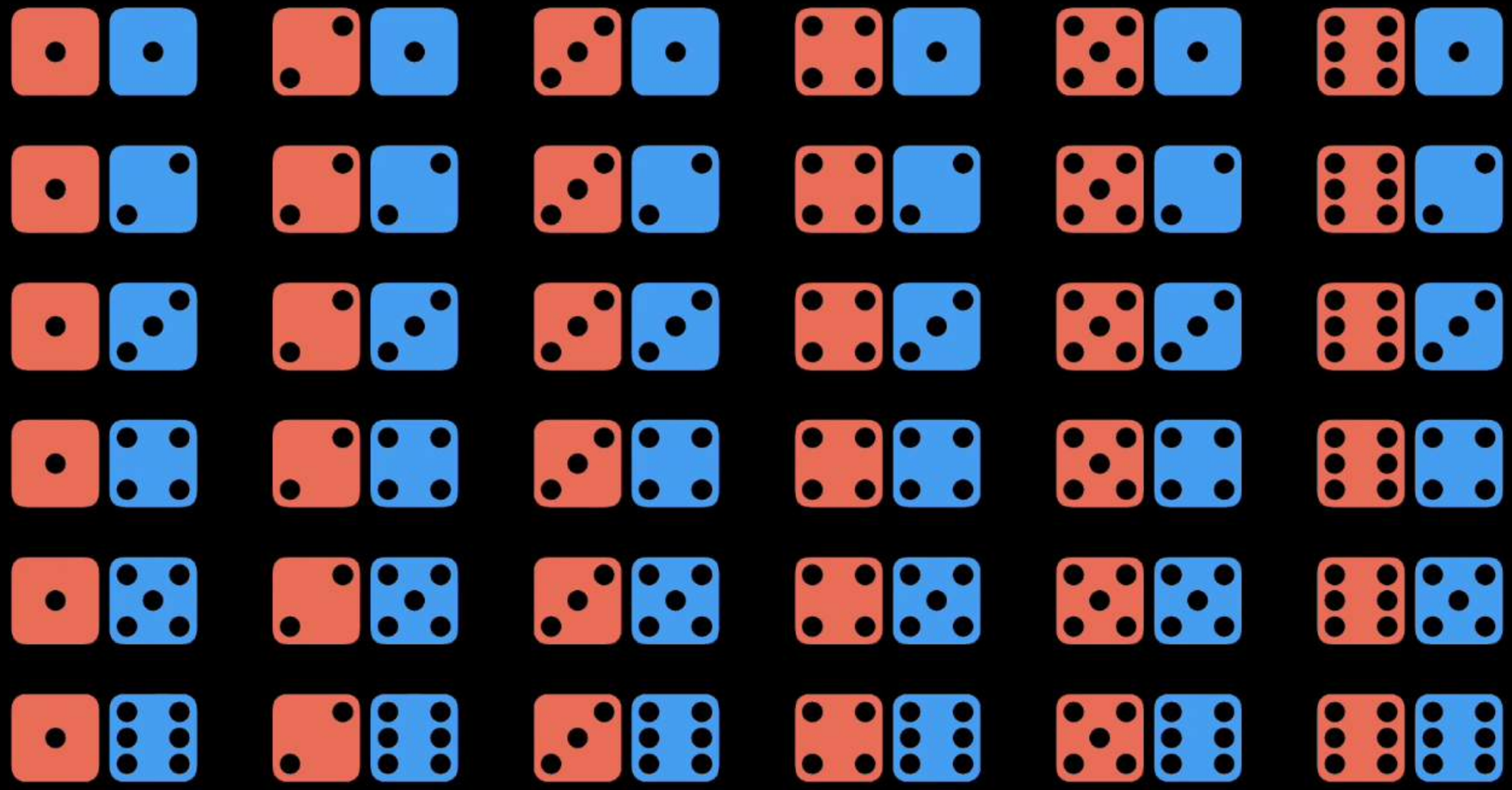
$$\sum_{\omega \in \Omega} P(\omega) = 1$$



$$\frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{6}$$

$P($  $) = \frac{1}{6}$





2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12

2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12

$$P(sum\ to\ 12) =$$

2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
7	8	9	10	11	12

$$P(sum\ to\ 12) = \frac{1}{36}$$

$$P(sum\ to\ 7) = \frac{6}{36} = \frac{1}{6}$$

Probabilidade Incondicional

Probabilidade de um evento ocorrer sem levar em consideração a ocorrencia de outro evento

Probabilidade condicional

Probabilidade de um evento ocorrer, dado que outro evento já ocorreu

Probabilidade condicional

$$P(a \mid b)$$

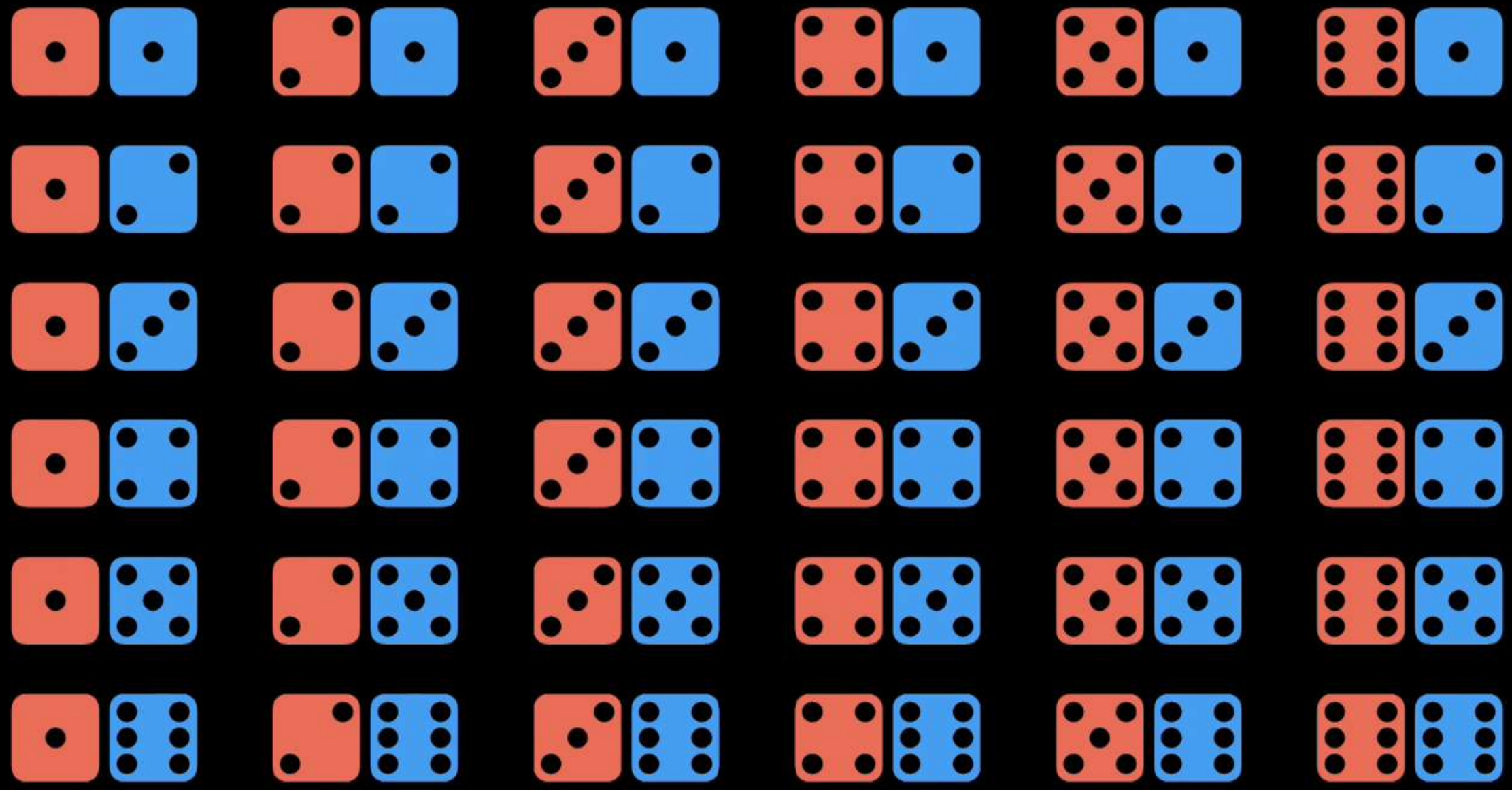
$$P(\text{chove hoje} \mid \text{choveu ontem})$$

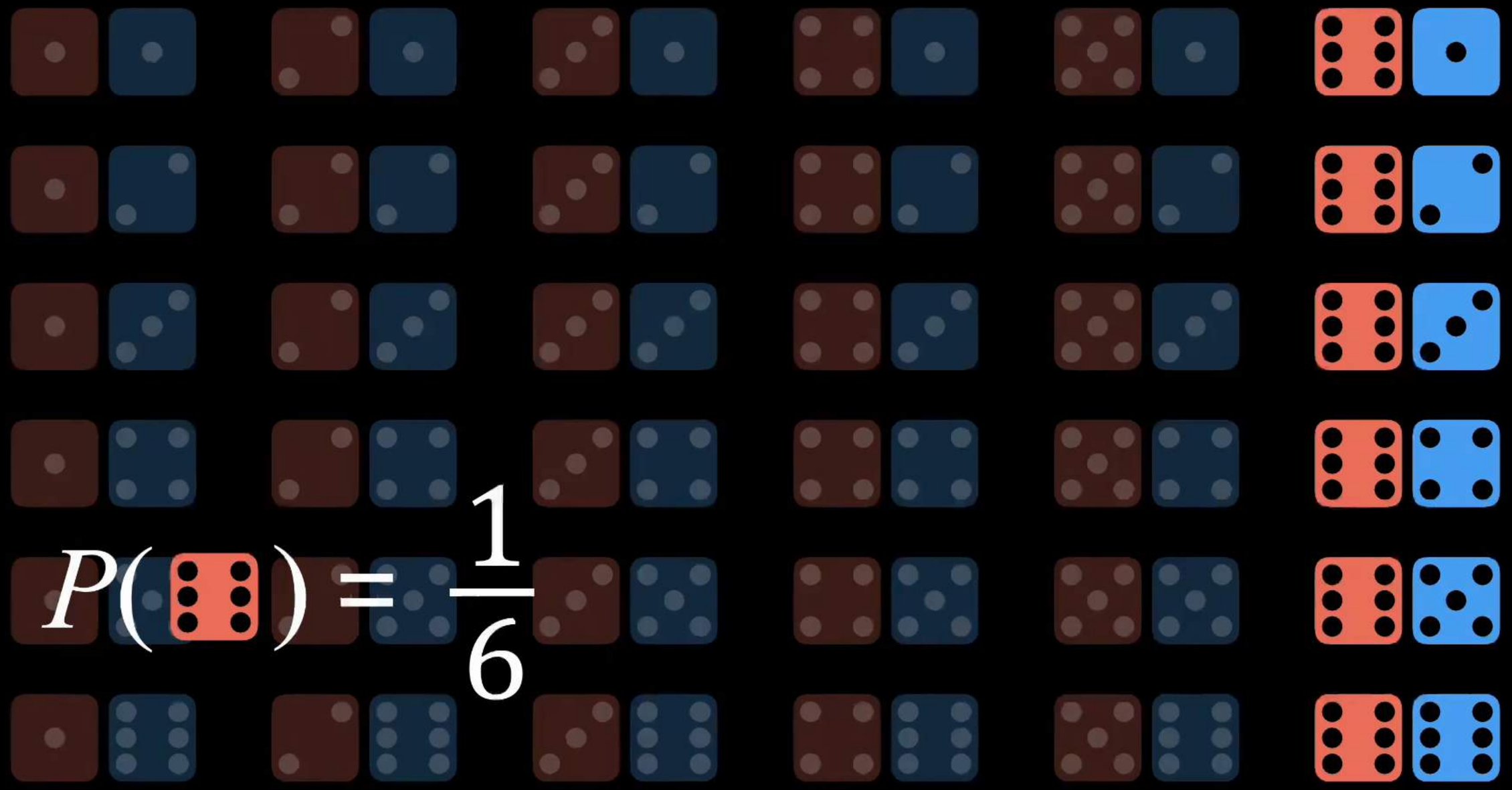
$$P(\text{mudança de trajeto} \mid \text{condição do tráfego})$$

$$P(\text{doença} \mid \text{resultado do exame})$$

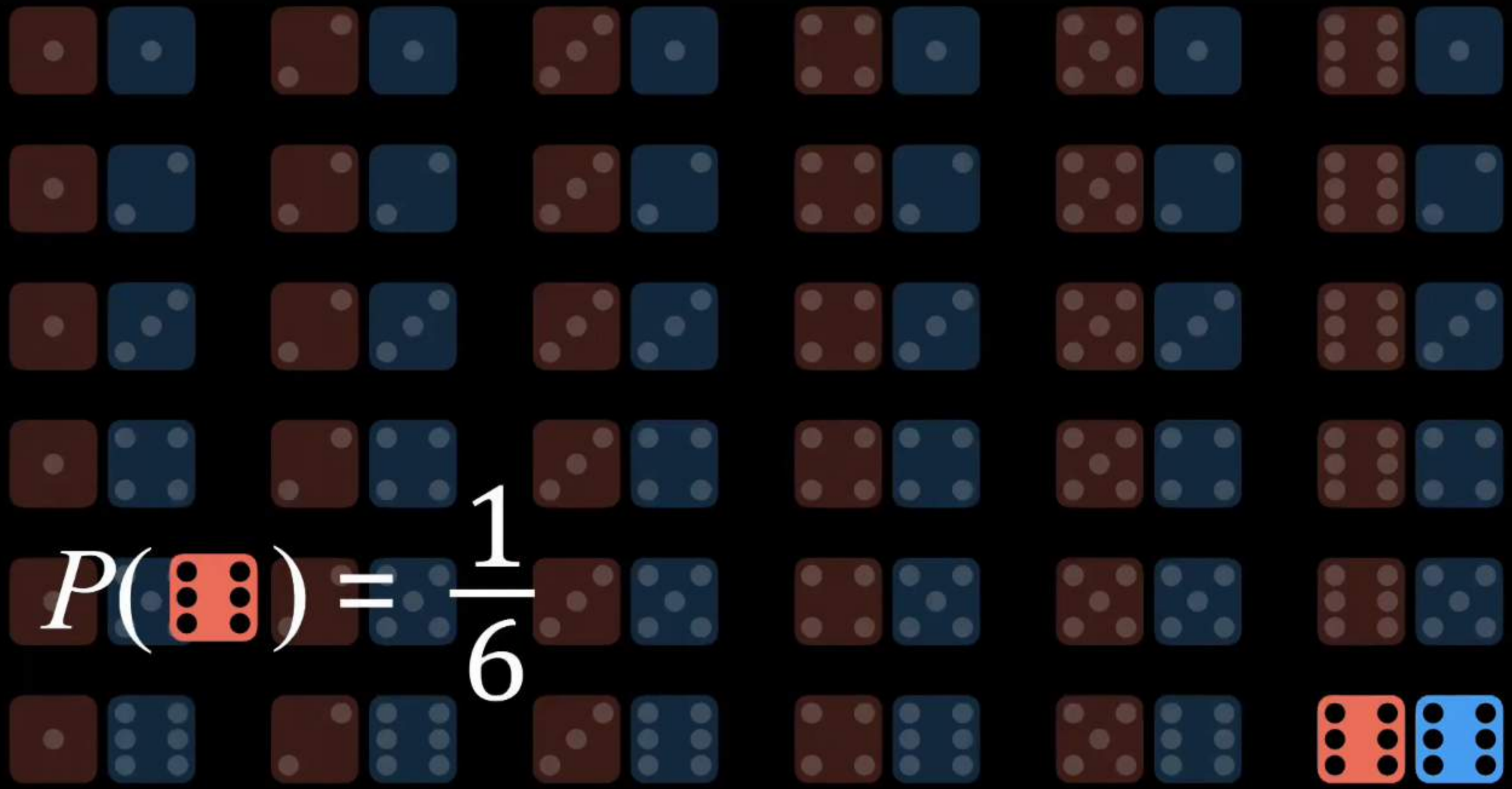
$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

$P(sum\ 12 \mid$ $)$





$$P(\text{⚅}) = \frac{1}{6}$$



$$P(\text{⚅}) = \frac{1}{6}$$

$P(sum\ 12 \wedge$ $) = \frac{1}{36}$

$P($ $) = \frac{1}{6}$

$$P(sum\ 12 \wedge \text{⚅}) = \frac{1}{36}$$

$$P(sum\ 12 \mid \text{⚅}) = \frac{1}{6}$$

$$P(\text{⚅}) = \frac{1}{6}$$

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

$$P(a \wedge b) = P(b)P(a \mid b)$$

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

$$P(a \wedge b) = P(b)P(a \mid b)$$

$$P(b \wedge a) = P(a)P(b \mid a)$$

Variável Aleatória

Variável que pode assumir diferentes valores com base em diferentes resultados de um experimento Variável aleatório

Variável Aleatória

Rolar um dado

$$\{1, 2, 3, 4, 5, 6\}$$

Variável Aleatória

Clima

{sol, nuvens, chuva, ventania, neve}

Variável Aleatória

Tráfego

$$\{nenhum, leve, pesado\}$$

Variável Aleatória

Vôo

{na horario, atrasado, cancelado}

Probabilidade Destributiva

$$P(Vôo = no\ horario) = 0.6$$

$$P(Vôo = atrasado) = 0.3$$

$$P(Vôo = cancelado) = 0.1$$

Probabilidade Destributiva

$$P(Vôo) = <0.6,\ 0.3,\ 0.1>$$

Independência

Quando a ocorrência de um evento não afeta a probabilidade da ocorrência de outro evento

Independência

$$P(a \wedge b) = P(a)P(b \mid a)$$

Independência

$$P(a \wedge b) = P(a)P(b)$$

Independência

$P($ $) = P($ $)P($ $)$

Independência

$P($ $) = P($ $)P($ $)$

$$= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Independência

Independência

$P($ $) \neq P($ $)P($ $)$

$$= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$$

Independência

$P($ $) \neq P($ $)P($ $)$

$$= \frac{1}{6} \cdot 0 = 0$$

Teorema de Bayes

Fundamentação Teorica

- Teoria da Probabilidade
- Teoria da Utilidade
- Teoria da Decisão

$$P(a \wedge b) = P(b)P(a|b)$$

$$P(b \wedge a) = P(a)P(b|a)$$

$$= P(b)P(a|b)$$

$$= P(a)P(b|a)$$

$$P(a)P(b|a) = P(b)P(a|b)$$

Teorema de Bayes

$$P(b|a) = \frac{P(b)P(a|b)}{P(a)}$$

Teorema de Bayes

$$P(b|a) = \frac{P(b)P(a|b)}{P(a)}$$

Dadas as nuvens pela manhã,
qual a probabilidade de chover à tarde?

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qual a probabilidade de chover à tarde?

- 80% das tardes chuvosas começam com manhãs nubladas.

Dadas as nuvens pela manhã,
qual a probabilidade de chover à tarde?

- 80% das tardes chuvosas começam com manhãs nubladas.
- 40% dos dias têm manhãs nubladas.

Dadas as nuvens pela manhã,
qual a probabilidade de chover à tarde?

- 80% das tardes chuvosas começam com manhãs nubladas.
- 40% dos dias têm manhãs nubladas.
- 10% dos dias têm tardes chuvosas.

$$P(chuva \mid nuvens) = \frac{P(nuvens \mid chuva)\, P(chuva)}{P(nuvens)}$$

$$P(chuva \mid nuvens) = \frac{P(nuvens \mid chuva)\, P(chuva)}{P(nuvens)}$$

- 80% das tardes chuvosas começam com manhãs nubladas.
- 40% dos dias têm manhãs nubladas.
- 10% dos dias têm tardes chuvosas.

$$P(chuva \mid nuvens) = \frac{P(nuvens \mid chuva)\, P(chuva)}{P(nuvens)}$$

$$= \frac{(.8)(.1)}{.4} = 0.2$$

- 80% das tardes chuvosas começam com manhãs nubladas.
- 40% dos dias têm manhãs nubladas.
- 10% dos dias têm tardes chuvosas.

Sabendo

$$P(\text{manhã de nuvens} \mid \text{tarde chuva})$$

a gente pode calcular

$$P(\text{tarde chuva} \mid \text{manhã de nuvens})$$

Sabendo

$$P(\text{efeito visível} \mid \text{causa desconhecida})$$

a gente pode calcular

$$P(\text{causa desconhecida} \mid \text{efeito visível})$$

Sabendo

$$P(\text{resultado teste medico} \mid \text{doença})$$

a gente pode calcular

$$P(\text{doença} \mid \text{resultado teste medico})$$

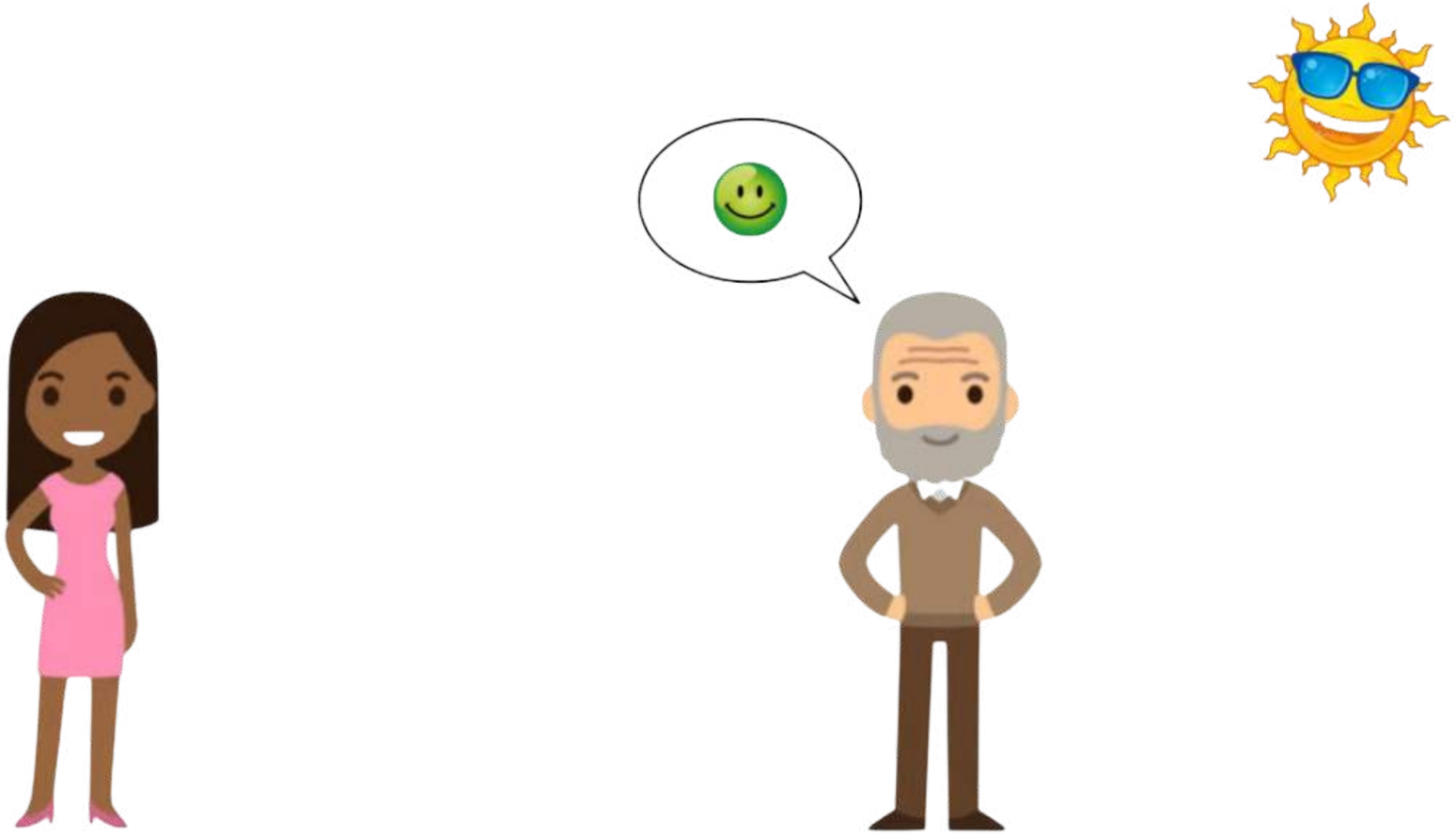
Sabendo

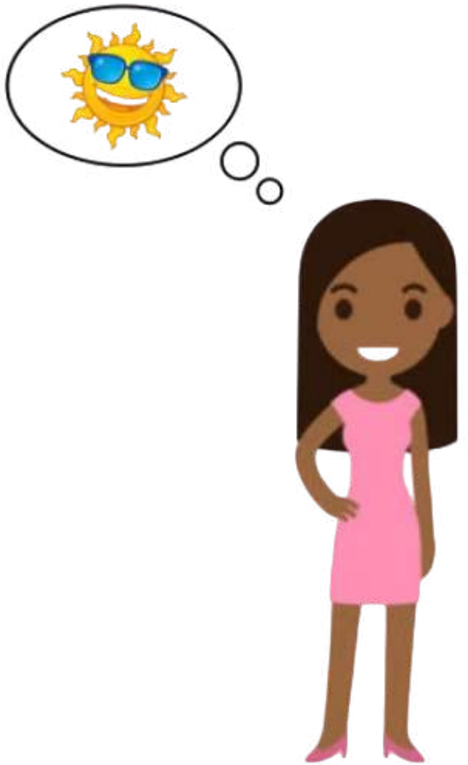
$$P(\textit{marca d'água visível} \mid \textit{nota falsa})$$

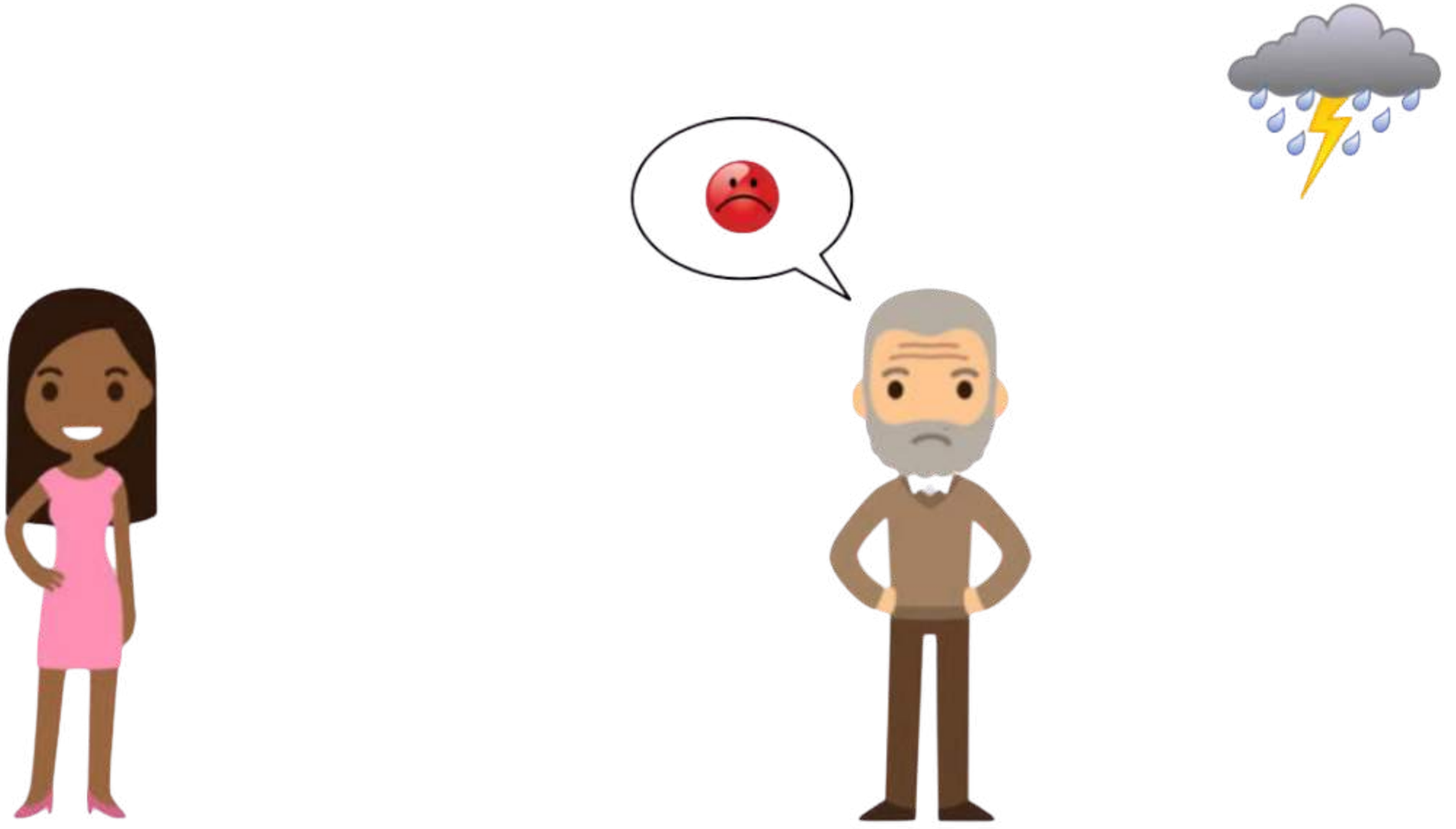
a gente pode calcular

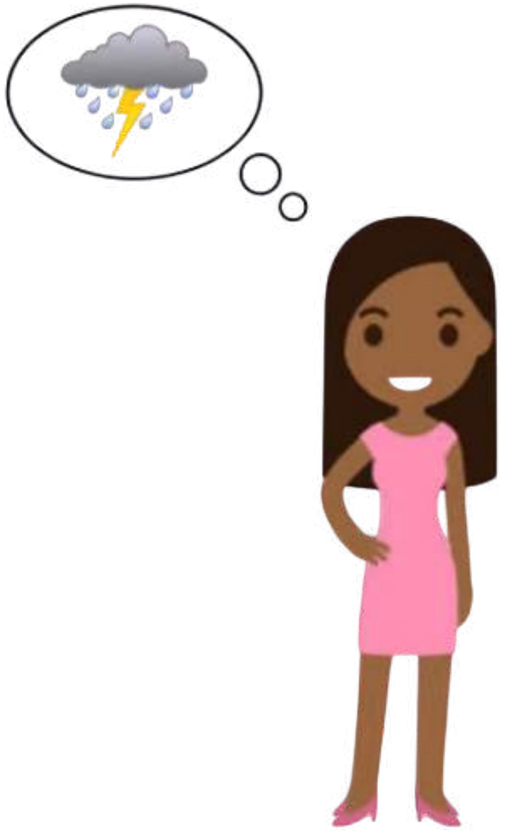
$$P(\textit{nota falsa} \mid \textit{marca d'água visível})$$

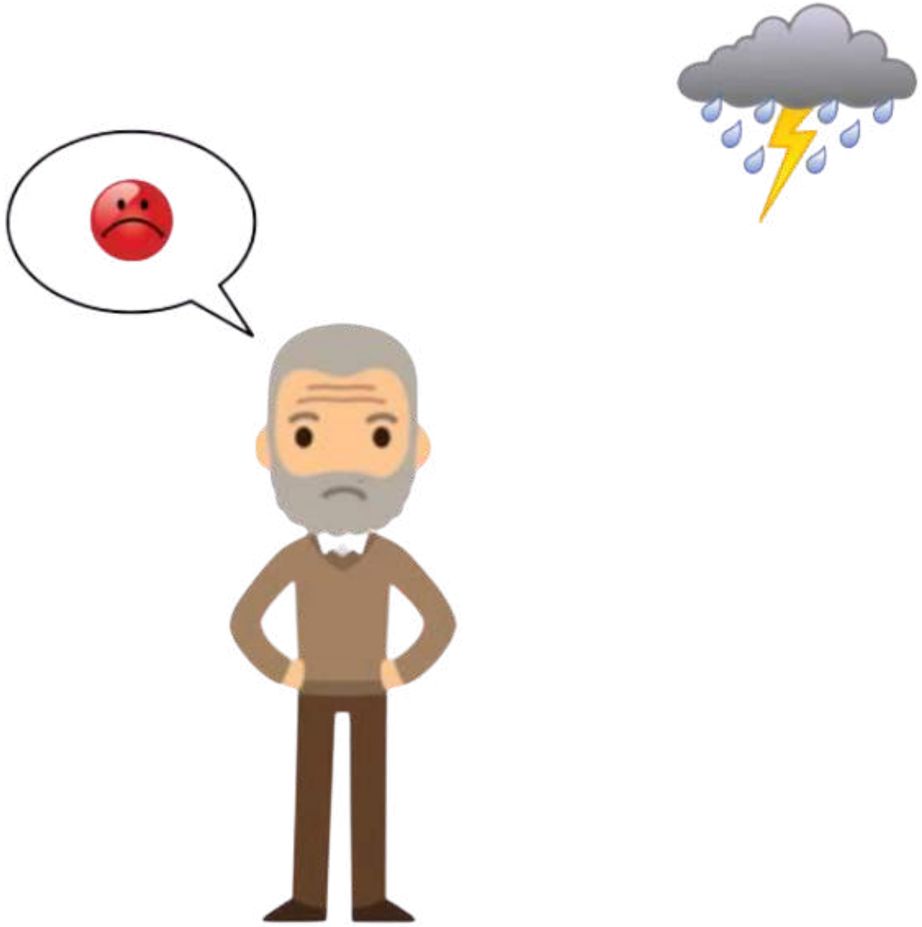
Teorema de Bayes +











Tempo

Tempo

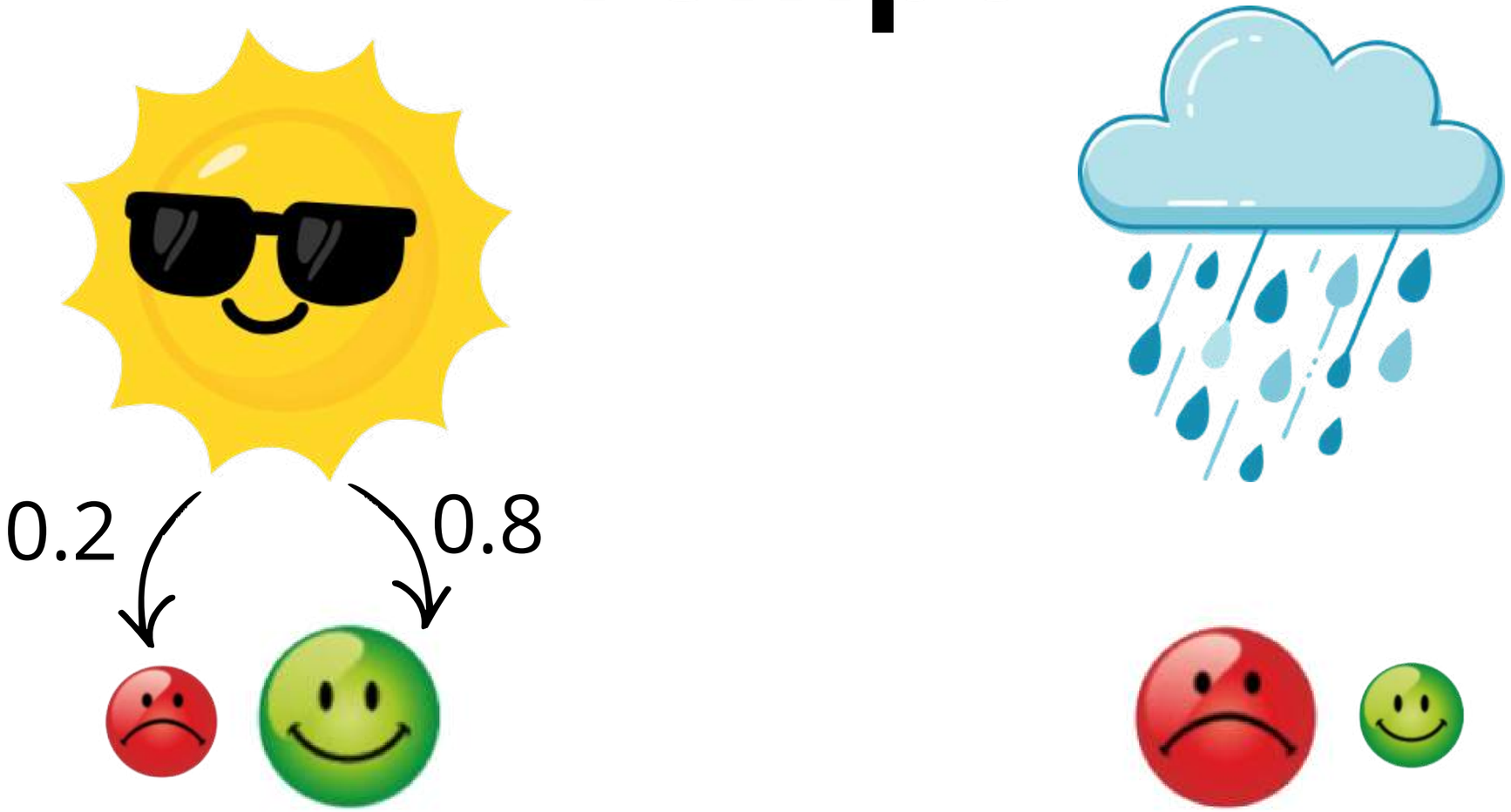
Tempo

Tempo

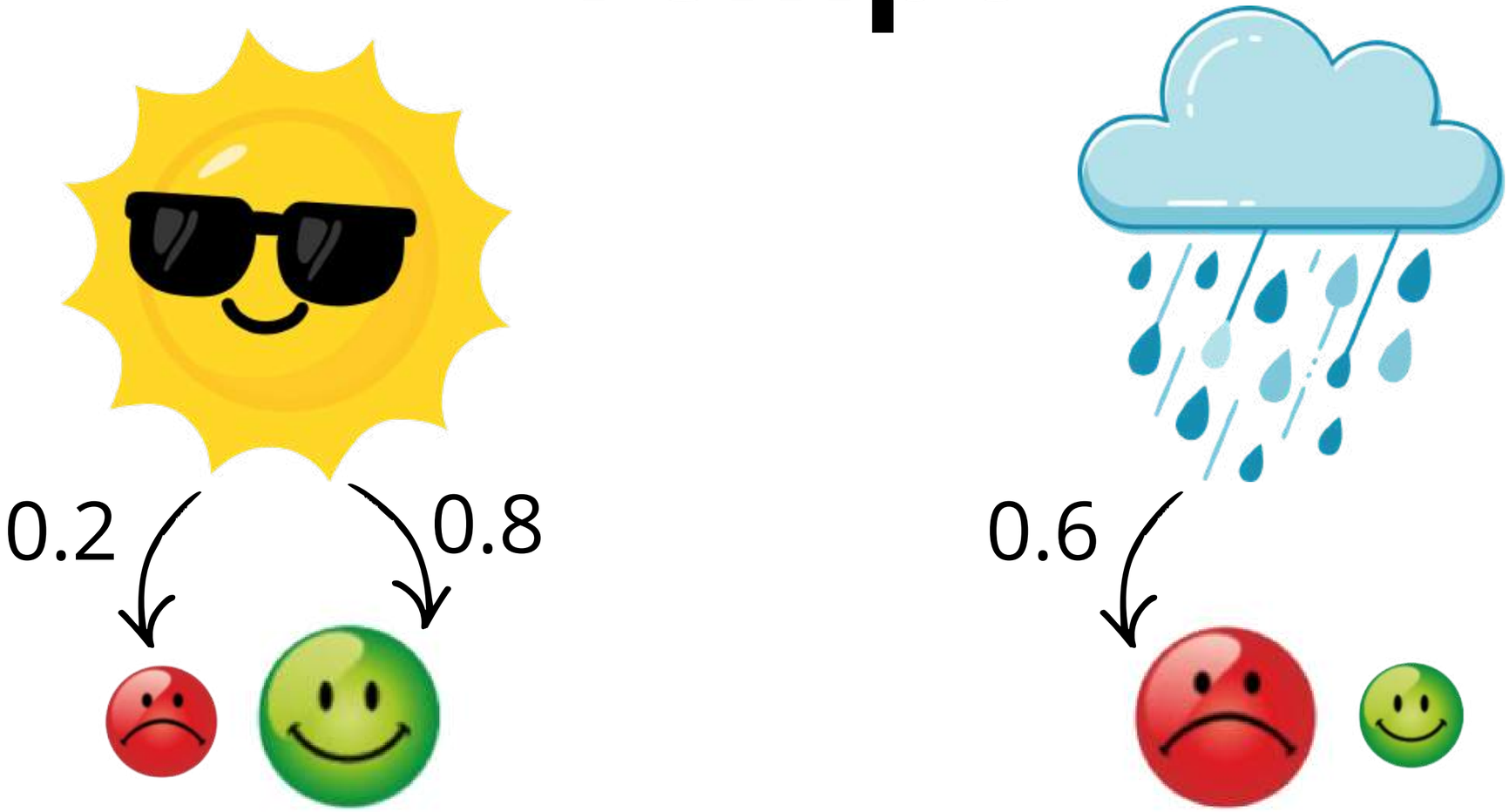
Tempo

0.8

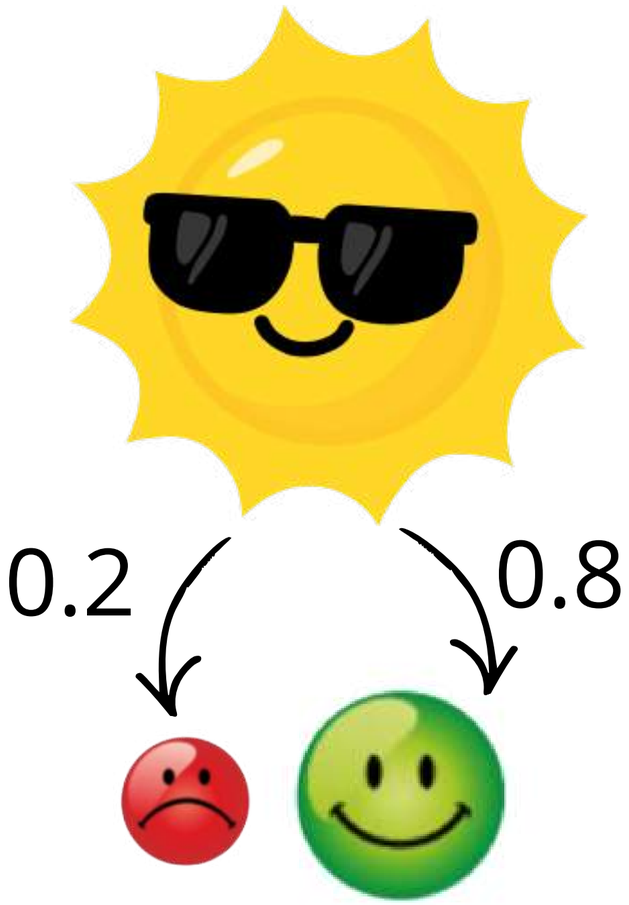
Tempo

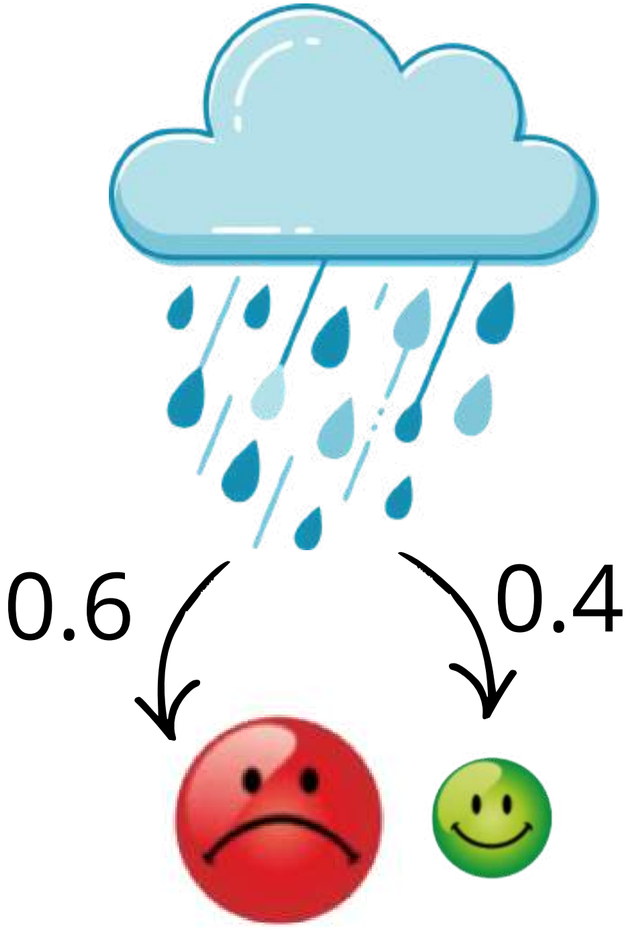


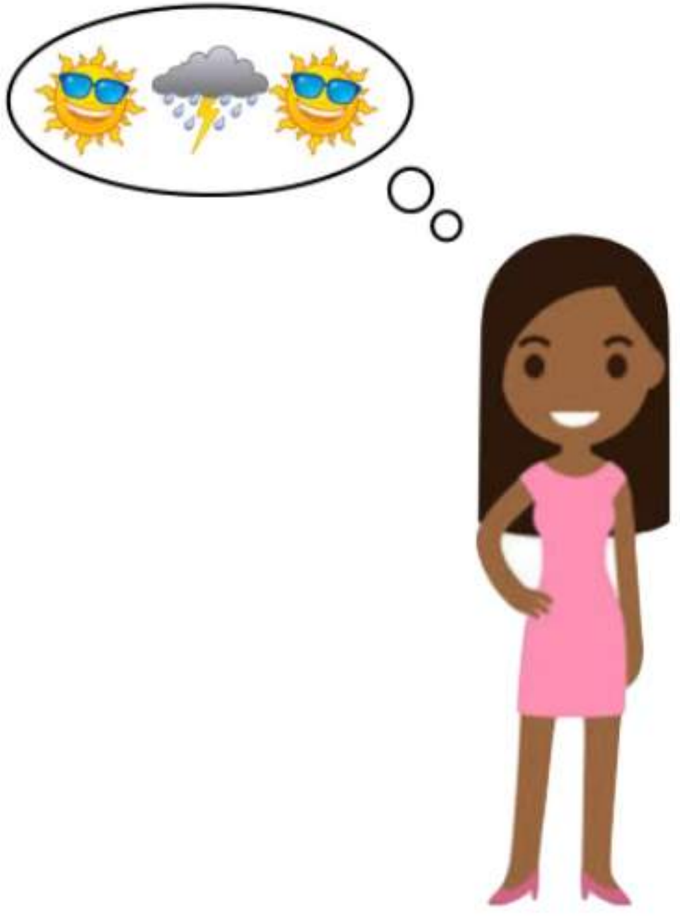
Tempo

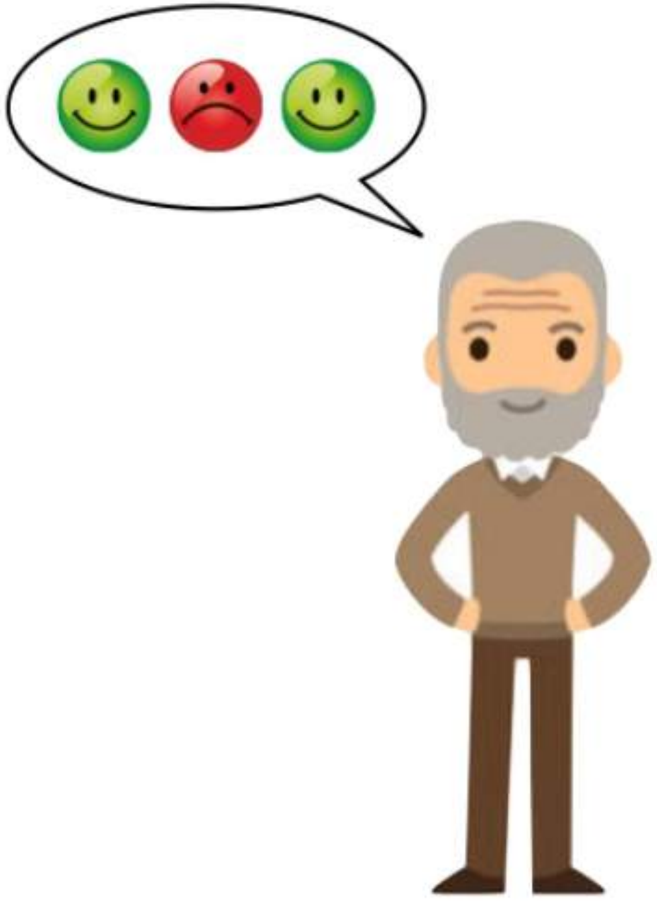


Tempo









Tempo

Segunda	Terça	Quarta	Quinta	Sexta	Sábado

Tempo

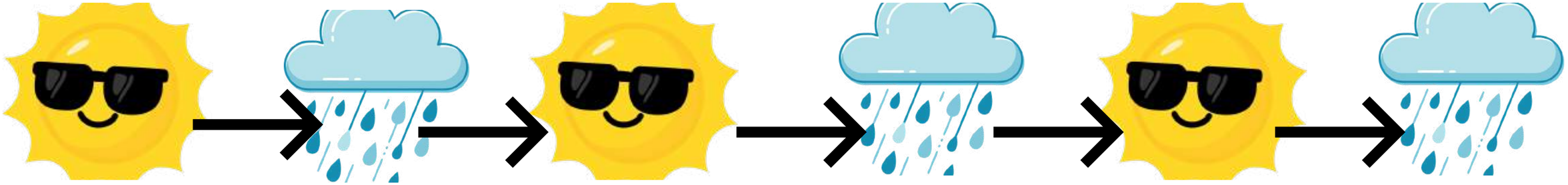
Segunda	Terça	Quarta	Quinta	Sexta	Sábado

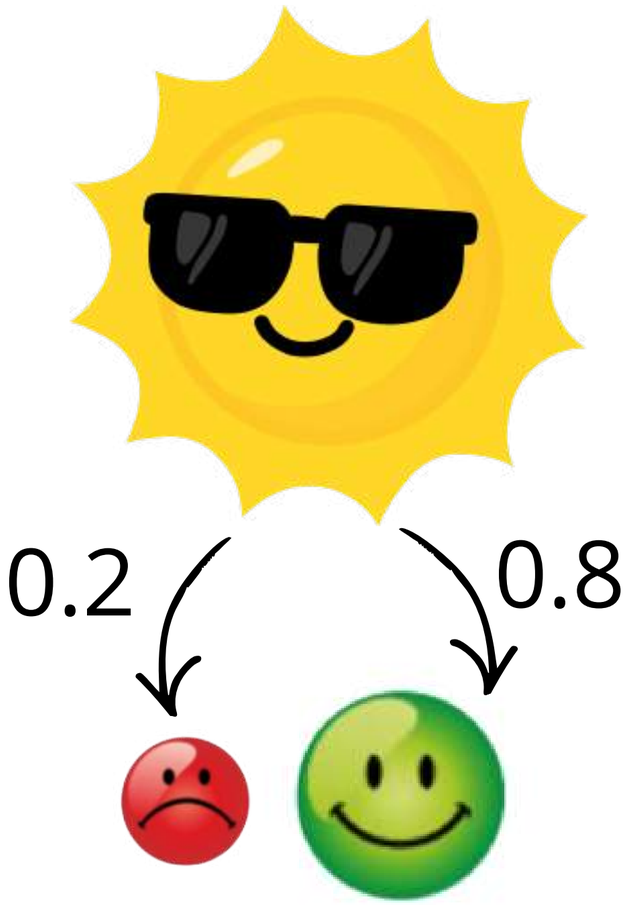
Tempo

Segunda	Terça	Quarta	Quinta	Sexta	Sábado

Tempo

Segunda	Terça	Quarta	Quinta	Sexta	Sábado

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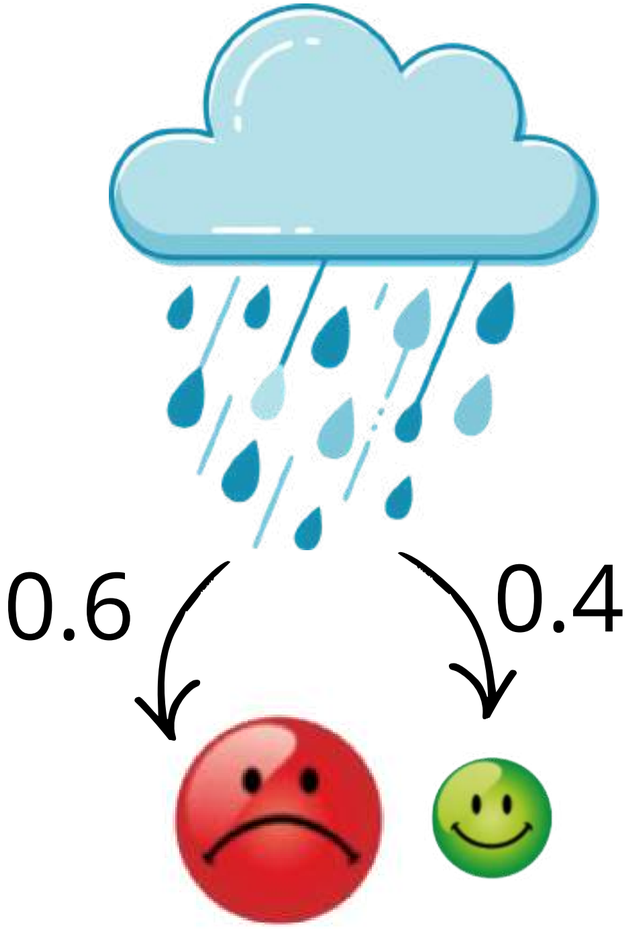
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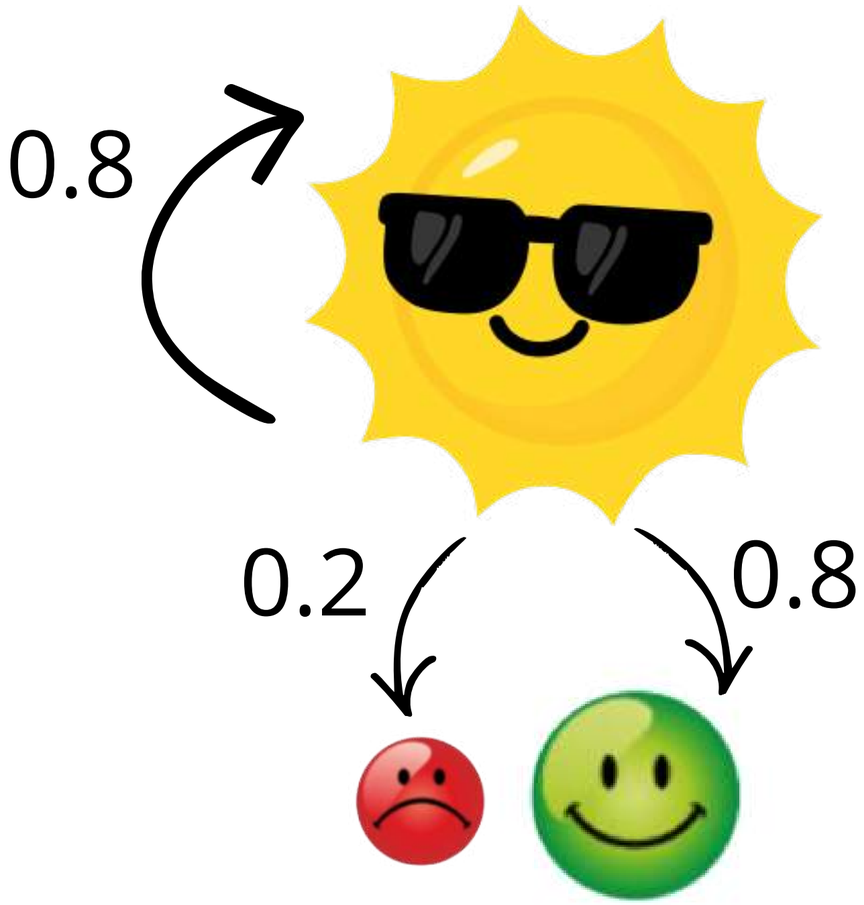
0.4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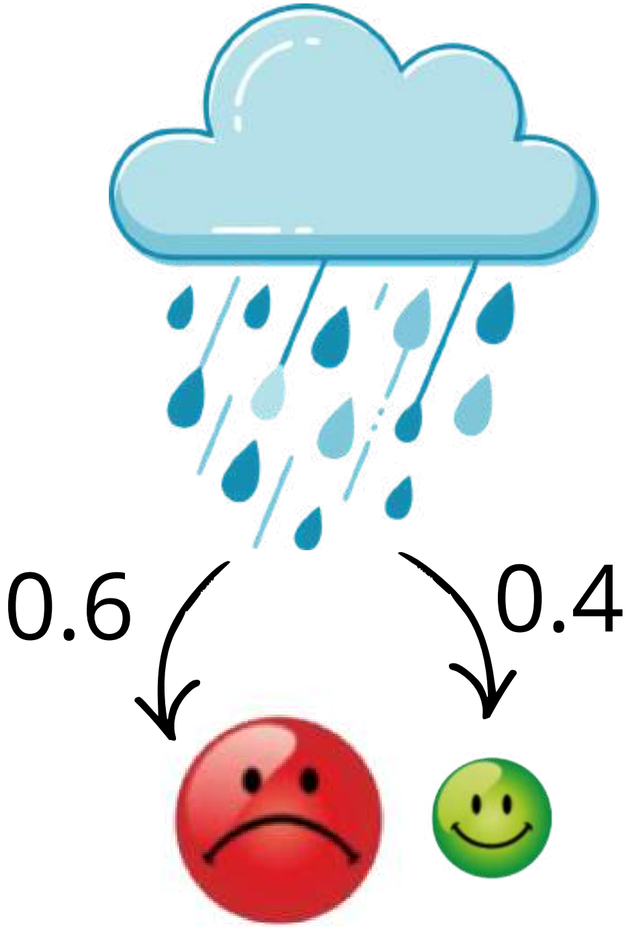
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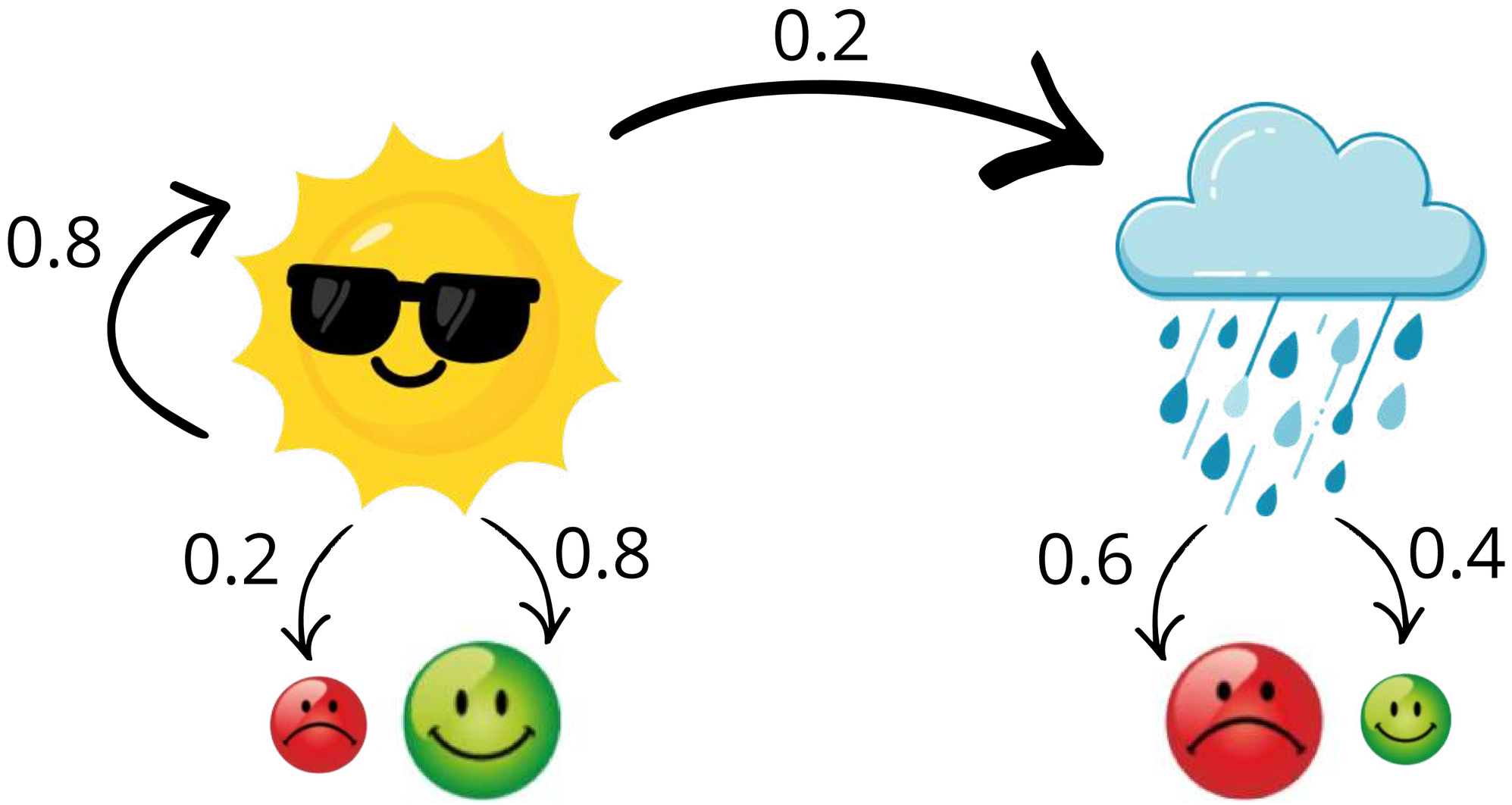
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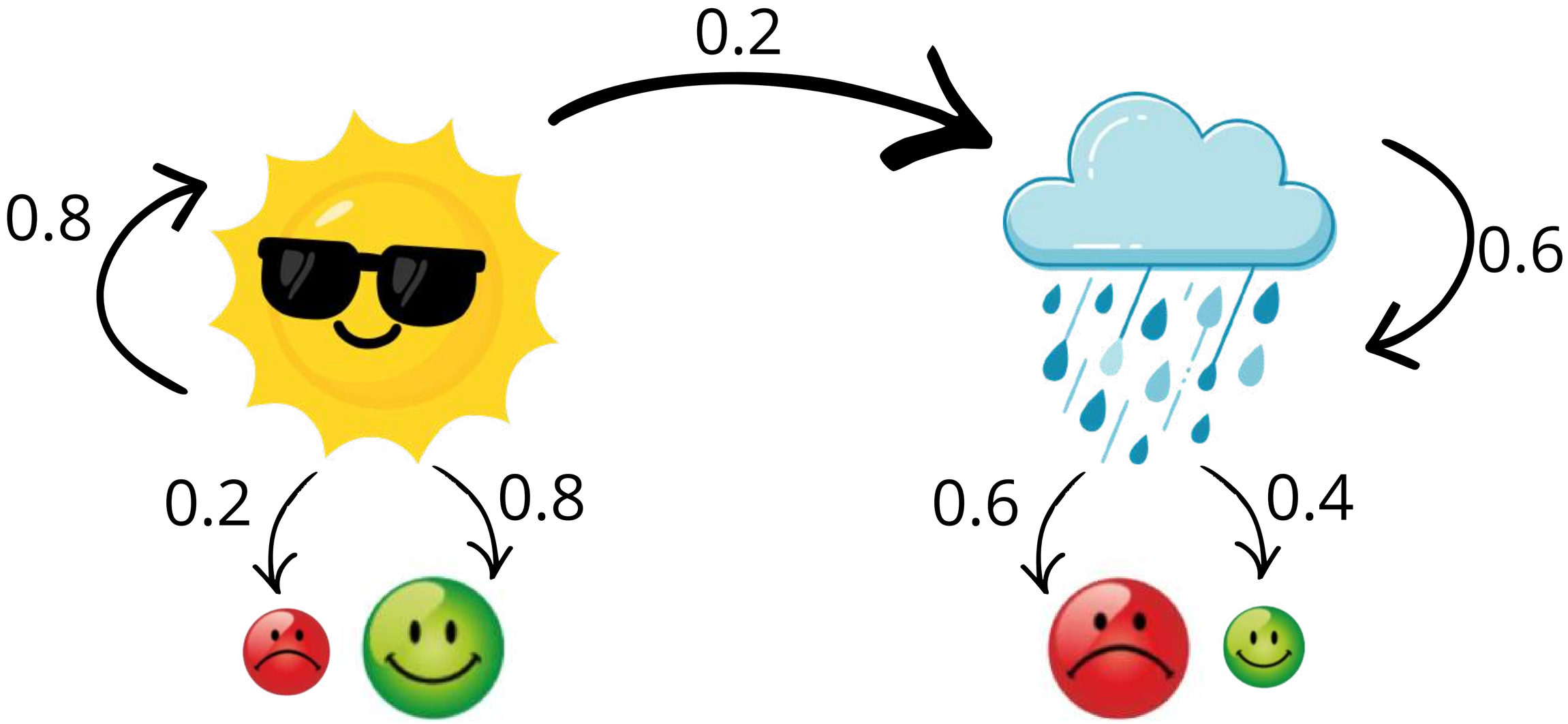
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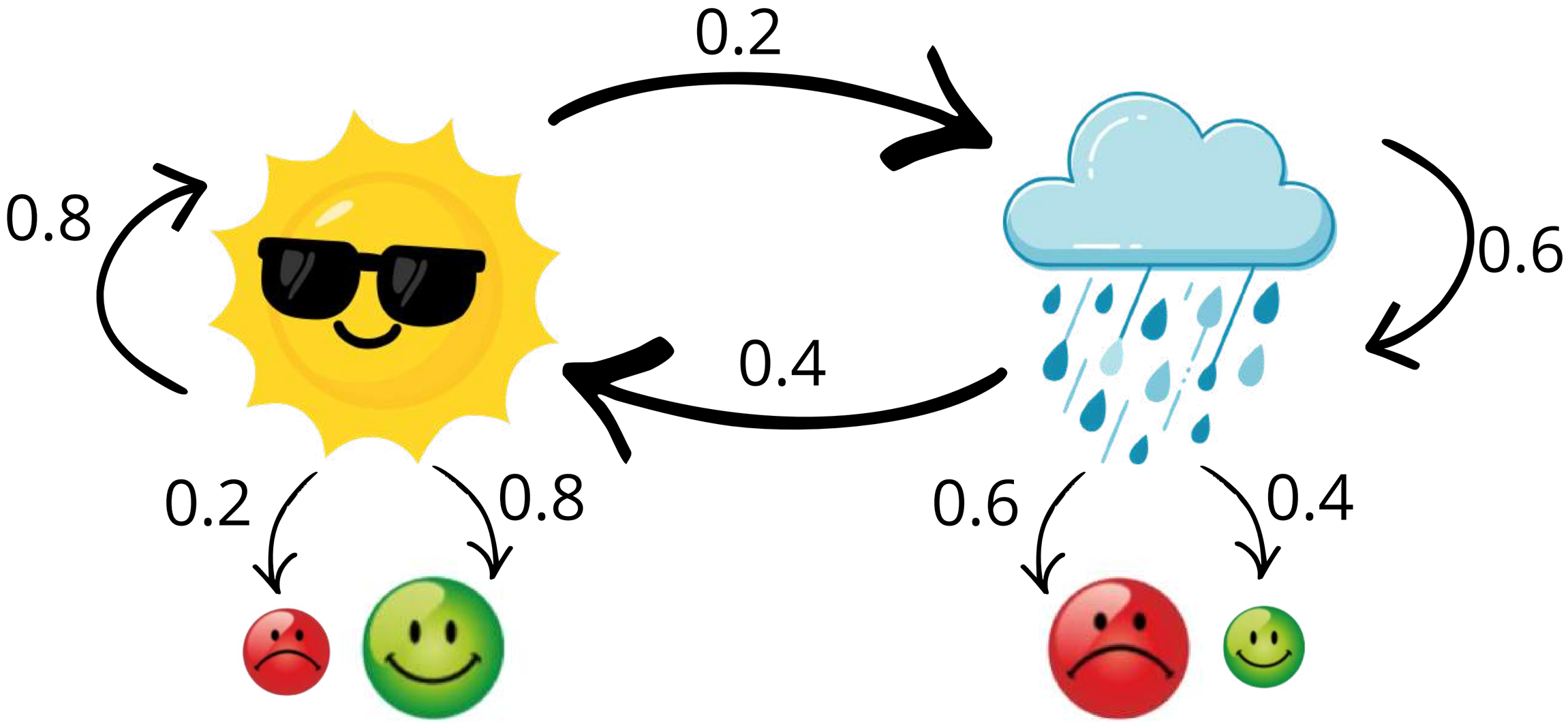
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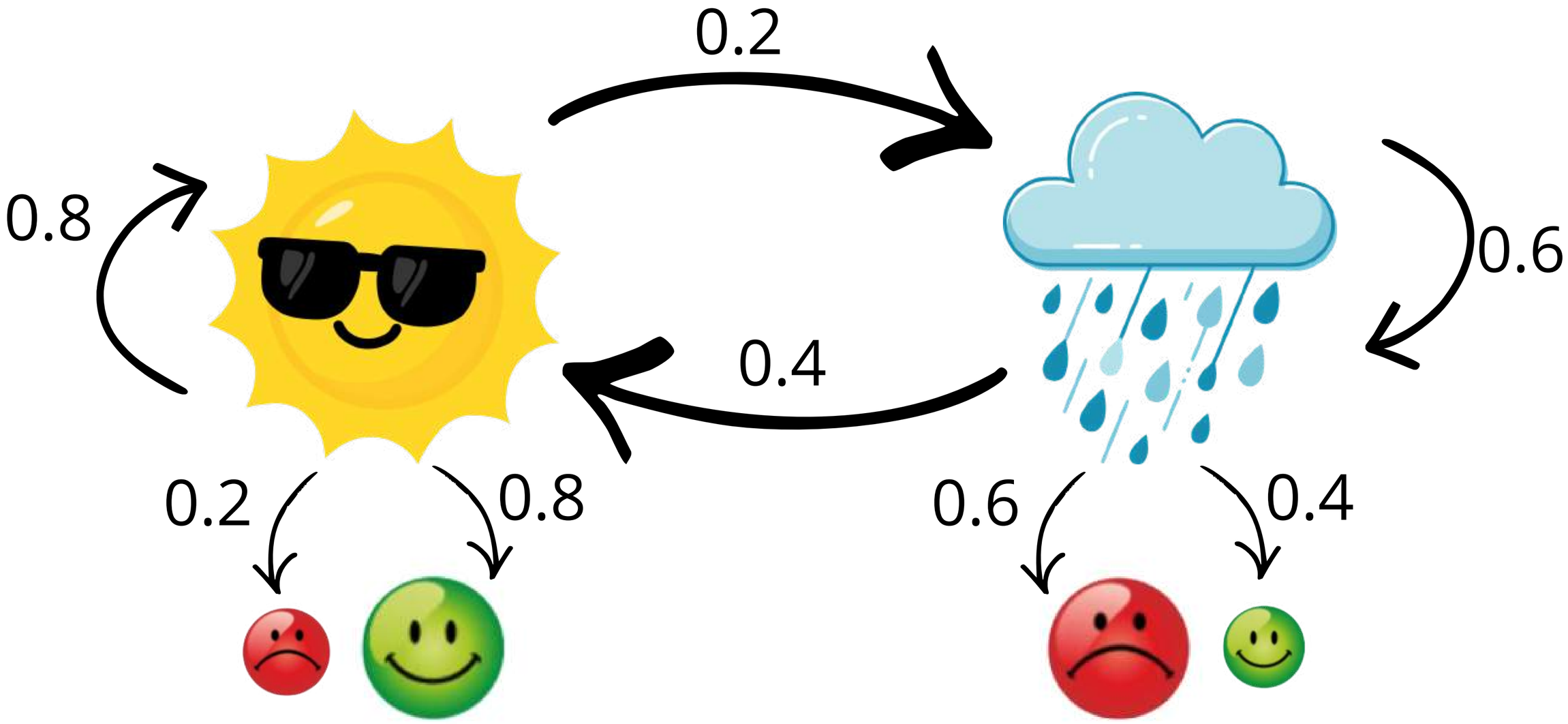
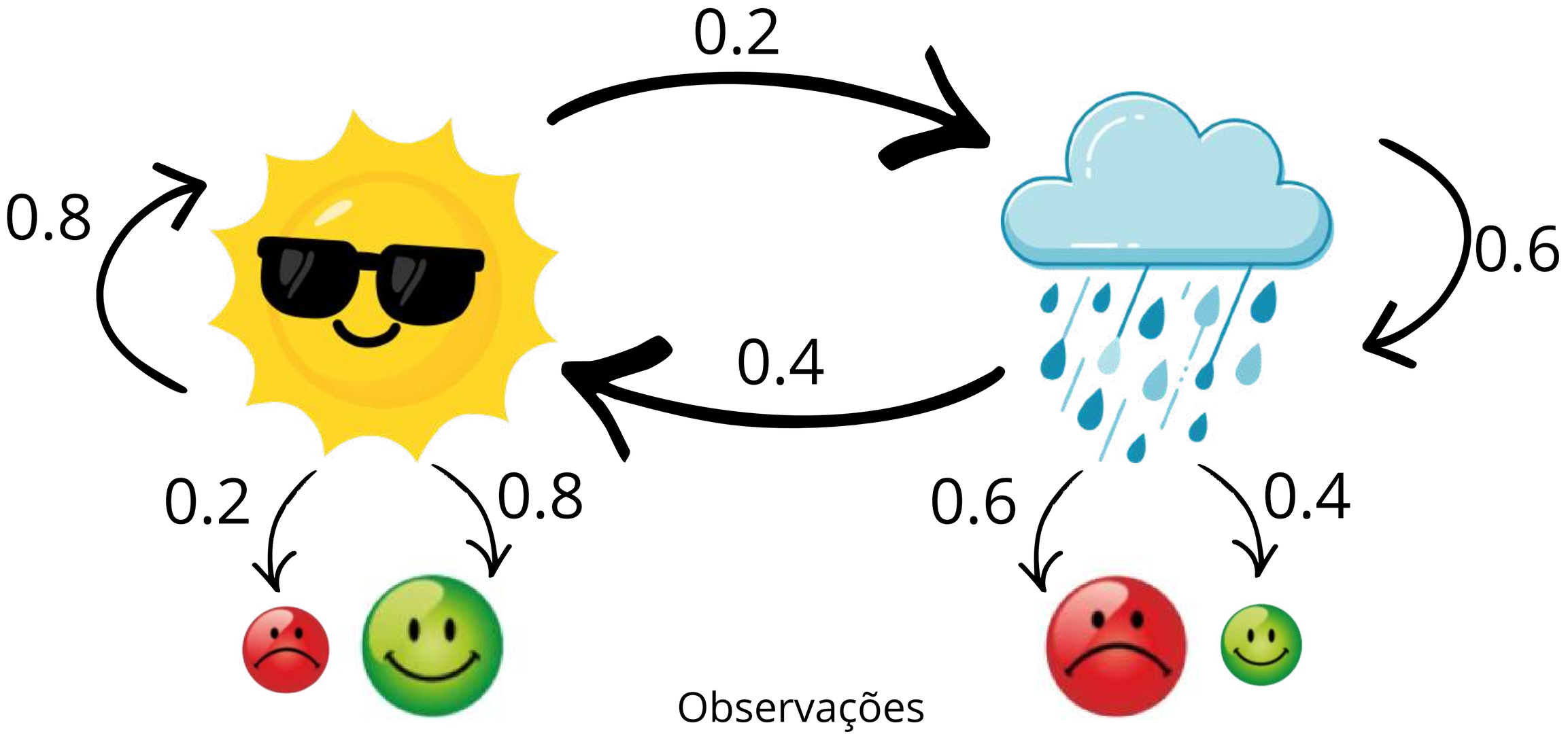
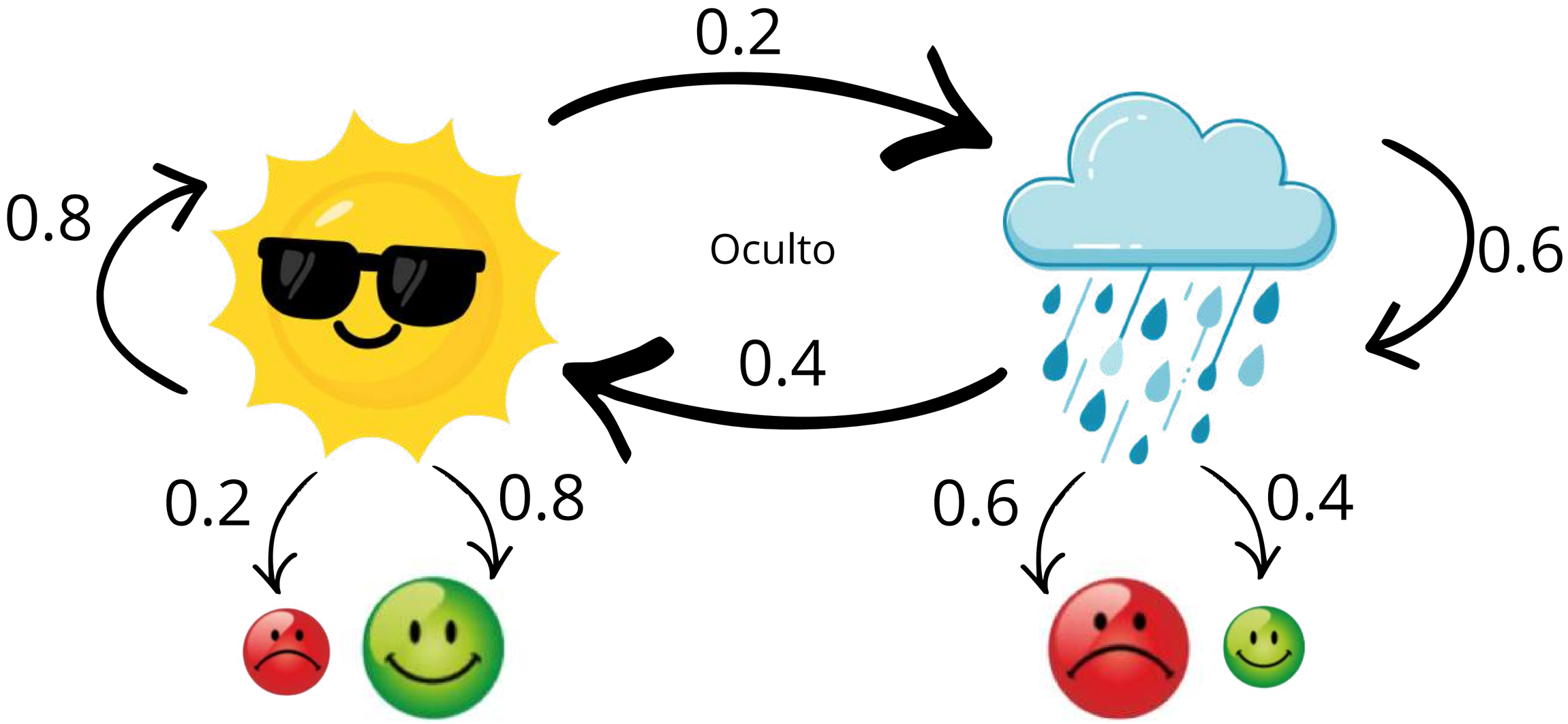


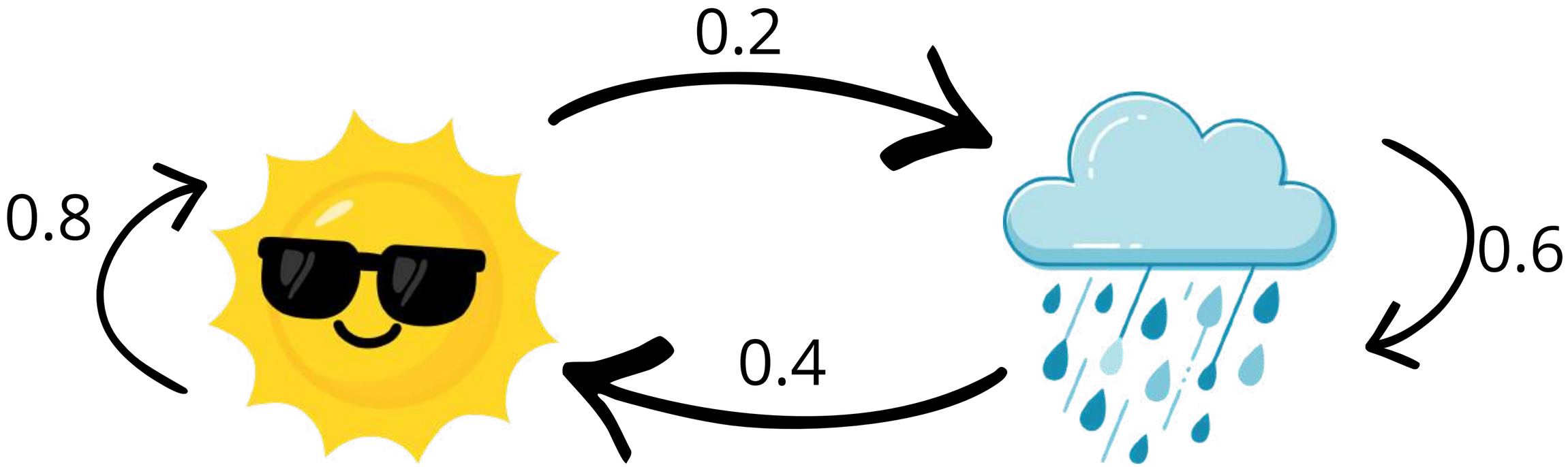
Modelo Oculto de Markov



Modelo Oculto de Markov



Modelo Oculto de Markov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0.8
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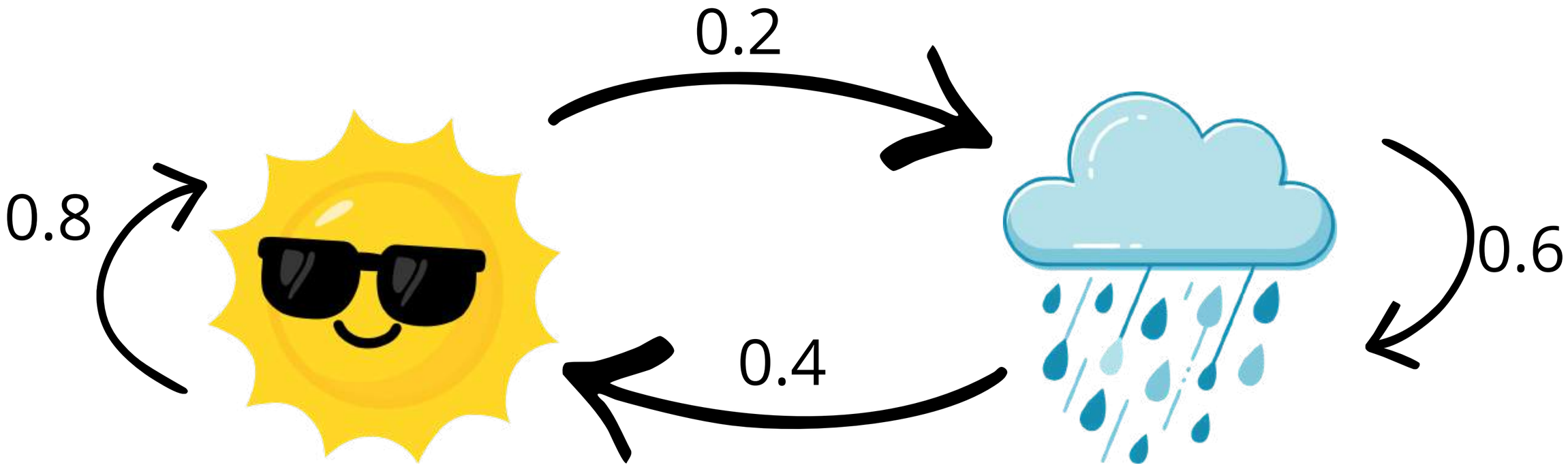
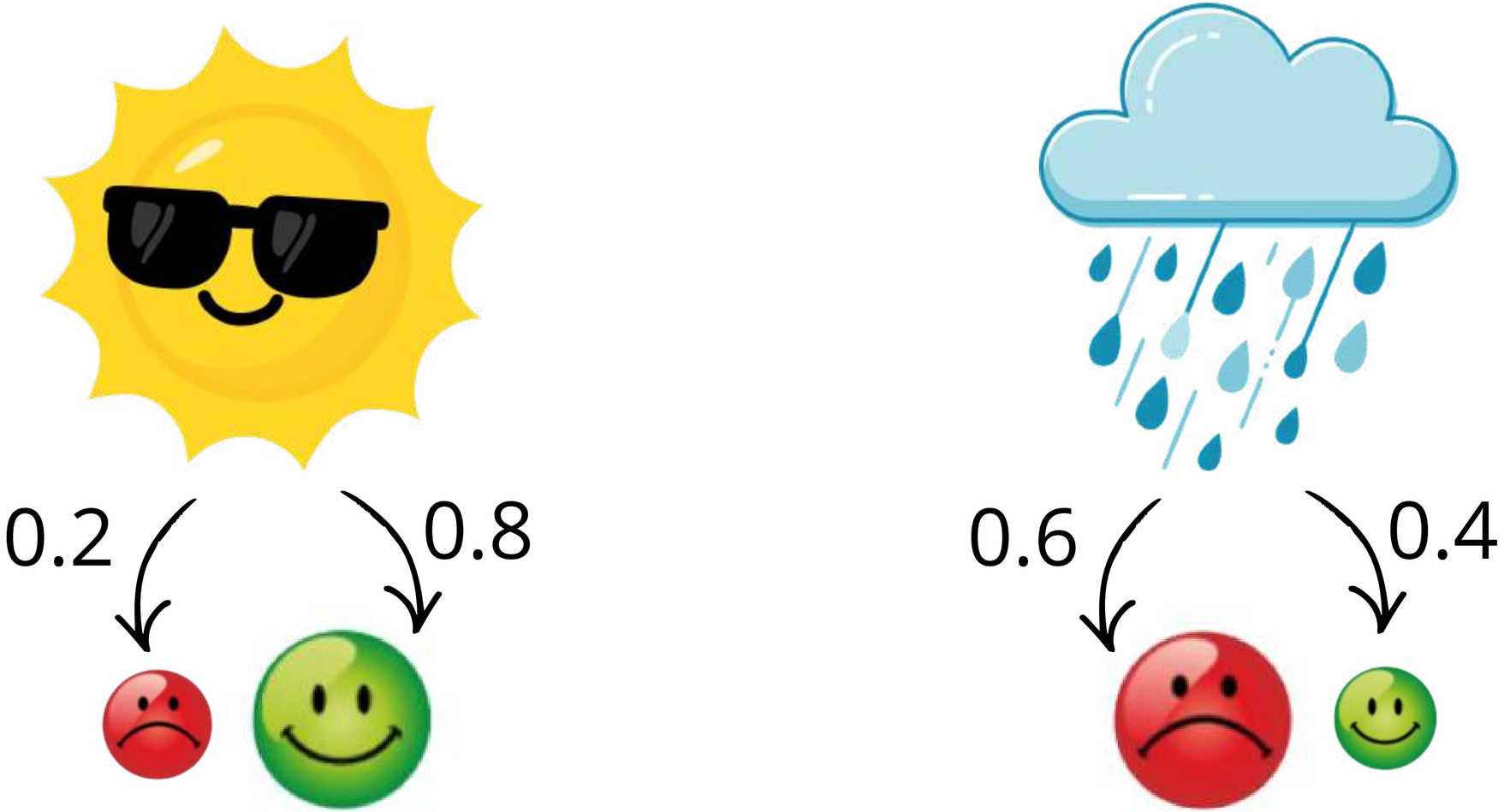


Probabilidades de Transição



Probabilidade de Emissão

Tomada de decisão na IA